

G. Lejeune Dirichlet

Werke, Band II, XXXVII

Correspondence between G. Lejeune Dirichlet and Leopold Kronecker

communicated by Ernst Schering

Complete English reading translation with mathematical notation retained in type

Introductory note by Ernst Schering

The following correspondence between G. Lejeune Dirichlet and Leopold Kronecker was communicated by Ernst Schering. Schering explains that, apart from the first letter, the letters belong to the period of Dirichlet's stay in Göttingen, and that this local connection gives them a special interest. Since the letters also have considerable historical importance for mathematical science, he asked Kronecker's permission to print them, together with some letters addressed by Kronecker to Dirichlet to which Dirichlet's replies refer. Schering knew that these Kronecker letters had been returned to their author by Dirichlet's son after Dirichlet's death. Kronecker granted permission and also allowed the letters addressed by him to Dirichlet, with Dirichlet's replies, to be handed over to the University Library at Göttingen.

I. Dirichlet to Kronecker

I have, dear friend, made myself guilty of such great negligence toward you that I can almost despair of obtaining your forgiveness. Being a young man, you will hardly find credible what I can put forward in my excuse: you can scarcely form an idea how much, in the middle age into which I entered some time ago, one lacks the necessary flexibility for quickly finding one's way into matters with which one is not already somewhat familiar. When your first letter arrived I was so overloaded with thirteen lectures a week and with other business that the little time I could devote to studying your interesting communication did not suffice to make your investigations clear to me.

I hoped that during the holidays I would be able to penetrate better into the nature of your work. My preliminary efforts had at least convinced me that in it you had taken up a much-treated subject from a new and, as it seemed to me, very happily chosen side. In this hope I was unfortunately disappointed, for immediately after the beginning of the holidays I was visited by an attack of influenza that made me incapable of any exertion for more than three weeks. Now, after having in recent days come nearer, I believe, to your work, I hasten to tell you that it seems to me very desirable that you publish a short note on your investigations, since your continuing ill health unfortunately prevents you for the present from thinking of a full redaction of them. If that note is so arranged that every mathematician not foreign to the subject can understand it completely, and if it does not exceed the limits set for articles admitted into the Academy's Monthly Report, I shall gladly present it to the Academy and move that it be admitted into the Report.

The rest verbally. I hope that your presence here will not fall, or at least not

wholly, in the period from 12 to 22 May, during which I accompany my wife, who is to visit Carlsbad. With my wife's and my own best compliments to you and to your wife, I call myself, your faithfully devoted Dirichlet.

Berlin, 3 May 1853.

Schering's notes. The two letters from Kronecker to which Dirichlet refers in this letter were not found in Dirichlet's papers. Their scientific content is reproduced in the note communicated by Dirichlet to the Berlin Academy in June 1853 and printed in the corresponding Monthly Report. Kronecker further informed Schering that at the end of May 1853, on a journey to Paris for consultation with a physician, he passed through Berlin and personally handed Dirichlet the manuscript of that note.

II. Kronecker to Dirichlet

Most honoured Professor, I have heard with great joy from many sides that you are very pleased with your new home and have every reason to be satisfied with the change, although that change has brought us here so grave a loss. I hardly need tell you how deeply I am involved in that loss and how keenly I feel it, for you know yourself how infinitely much I owe you, and you know also that the main purpose of my move to Berlin was to be nearer to you.

I had intended to visit you this month, so that for at least a few days I might have the long-missed pleasure of personal intercourse with you. But Borchardt told me that you are already going to Paris in a few days, probably to remain there four to six weeks; I shall therefore postpone the execution of that plan until after your return, provided that you still maintain the permission you earlier gave me orally to visit you. I shall learn in due time, if not directly from you then indirectly through Borchardt or other acquaintances here, when you are back in Göttingen.

For that time I could of course reserve all other communications; but who knows what may intrude between intention and execution. Since I still believe myself assured of your old interest, I shall add a few words about my mathematical life during the past winter. I worked quite steadily during the whole time since my return from the seaside, and the results obtained have for the most part rewarded my efforts. I did not solve the principal question to which I came about a year ago, but I gained an insight into it that is very valuable to me. Moreover, without that solution I have attained the original purpose for whose sake I believed that I had to solve that question, so completely that nothing remains to be desired in that respect.

I have found a method for deriving all the properties of solvable equations of prime degree; in simplicity and rigour it should satisfy every just demand. The method requires no higher standpoint of mathematical conception than the problem itself, which is disposed of by it. I shall devote the summer to a careful working out of these matters. The work will become fairly extensive, not because of obscurity but because very many individual results come out, and because I must not spare space in precisely designating certain quite new points of view, especially those missing in the Galois and Abel fragments, without which the required rigour cannot be observed. Some pleasing results—or more exactly, pleasing ways of viewing things—which I obtained while occupied with these methods, I shall soon publish in a small note, because in the complete papers now to be drafted these matters would come quite last and thus would not reach publication for about a

year.

In that note I shall also give the most general root of an Abelian integral equation represented as a complex number. I have now succeeded in putting it into so simple a form that the necessary preliminary number-theoretic remarks are only quite insignificant. While in this way I brought the investigation of the solvability of equations of prime degree completely to an end, I have also occupied myself in many ways with more general work: on the one hand by undertaking the investigation of solvability for degrees that are not prime, and on the other by no longer restricting the investigation of equations to solvability, but by extending it to the discovery of all properties that an equation can have at all.

In this latter direction I have admittedly only reached the point of seeing the whole great new field before me, that is, of knowing clearly what is to be sought; but in an investigation of such generality I do not think that insignificant. The complete solution of the precise problem I have set myself here will, if possible at all, only fall to my weak powers after a considerable time. Yet it is certainly worthwhile to expend time and effort on a single question that comprises the entire theory of algebraic equations in one unknown. I add only that I have solved this problem for the first degrees, including the seventh; that the difficulties begin only with degree seven; and that already in this special case I obtained results which, by their novelty and by their remarkable connection with complex number theory, seem very interesting to me.

The naturalness of these abstract algebraic investigations appeared to me in the noteworthy circumstance that, just as all solvable and Abelian equations occur in theories of transcendental analytic functions, so also all the “affects”-to use Jacobi’s expression-which I had found a priori to be possible actually occur in certain known equations of the seventh degree. As for the solvability of equations whose degree is an arbitrary composite number, I have also made substantial progress, though without having brought the matter to complete closure. My work on equations of prime degree will supply the simple points of view by which the algebraic and the number-theoretic are properly separated in this whole domain, and by which the clarity and security hitherto everywhere lacking are obtained.

You will surely approve that I now interrupt the general investigations just mentioned in order to begin the redaction of the material-considerable at least in extent, if not in content-which has accumulated through nearly three years’ work. I also want to finish at once a few very small papers to send to Liouville. You would greatly oblige me if, in Paris, you would occasionally use your weighty word to recommend their prompt acceptance in the Journal. In the first I shall explain the small point about Bernoulli numbers that I found more than ten years ago; in the second I shall give, in a form suitable for lectures on cyclotomy, an extremely simple way of determining the sign of

$$\sum e^{\frac{2k^2\pi i}{p}}.$$

In the third I shall give a very simple proof of the irreducibility of

$$1 + x + x^2 + \cdots + x^{p-1},$$

and the fourth will contain an essential shortening of a method used by Kummer in his paper on complex numbers. These, most esteemed Professor, are my good intentions for the next weeks. I postpone the working out of many other things I did this winter and earlier because I do not yet consider them ripe; in my work on

solvable equations I have learned that I must wait matters for a long time before they attain the maturity needed for publication.

I have surely put your patience as a reader to too hard a test, and therefore I shall finally close these lines, wishing you a very happy journey and recommending my wife and myself most warmly to you and to your wife. With most sincere reverence and attachment, your grateful pupil, Leopold Kronecker.

Berlin, 3 March 1856. I add Kummer's best greetings.

III. Kronecker to Dirichlet

Most honoured Professor, when I wrote to you on 3 March I had already considered the possibility that my intended trip to Göttingen might be prevented by something, but there was nothing then to fear directly. Since then my old ailment has again tormented me; my little son, ill throughout the preceding winter, needed a change of air as soon as possible; and so for several weeks we have been here in Kösen, from where I shall go to a sea-bath at the end of July in order to bring my own health as much as possible back on track. Since in this way I am again thrown back upon preserving myself in your memory in writing, you receive these few lines merely as an accompanying letter to the offprint of a little article, whose occasion I already described to you on 3 March. I hope that occasion will sufficiently excuse, in your eyes, the existence of the enclosed trifle, should you glance into it.

In that case I mention only that the numbers denoted by k on page 209 and their properties cost me more trouble than they look, and that the derivations and proofs of the results communicated about Abelian equations offered various difficulties precisely in the detailed working-out that was required. Moreover those numbers k are interesting in more than one respect, and there is no doubt that for certain enumerations of forms they play the same role as the numbers denoted by you by a and b in the theory of quadratic forms.

If, keeping the meanings assigned to the letters n and m in the little article, one considers in general forms of degree n in several variables, then those corresponding to the simplest complex numbers can be defined as follows: the forms must be decomposable into linear factors, and the cyclic functions of these linear factors themselves must already be rational functions of the variables, that is, have rational coefficients. Such forms of degree n with determinant m are nothing other than the norm forms, and their associates, of complex numbers built from the periods $\omega(\varrho)$; and it is in the number of such forms that the k 's must appear.

In this way one has a simple explanation of the forms that are plainly the nearest at hand, and none of the accidental peculiarities occurs which Eisenstein had to call upon in explaining the cyclotomic forms of the third degree in three variables. As for my work, I have advanced only little in my main occupation, because I had to occupy myself with several small elaborations. But in hopeful prospects, in what I can prove, I have often run ahead and gained views of the nature of equations which, if they stand up, should afford an entirely new and substantial interest. Yet because the field on which I am now working is wholly new and because its complexity is scarcely to be mastered, I believe that I shall need years of work before I arrive at strict results. Therefore, honoured Professor, do not expect something from me in this respect very soon, especially since I shall devote a good part of my time to the most careful elementary treatment of my older results. And

I have nothing to miss, provided that my health improves with the years rather than becomes worse.

One thing more. In connection with a small proposition about integral equations, which will probably be printed in the first number of the Journal appearing under Borchardt's name, I returned several times to your note on complex units in the Monthly Report of the Academy. Allow me to say that your presentation there, as I believe, can lead to misunderstandings, and that in my opinion it would be better partly to begin the matter from the end. Suppose—as you show can always be found—one has a system of units

$$\Phi_1(\alpha), \quad \Phi_2(\alpha), \quad \dots, \quad \Phi_{h-1}(\alpha),$$

for which, if α' is conjugate to α ,

$$\Phi_k(\alpha)\Phi_k(\alpha') = a_k, \quad \log a_k = A_k,$$

and if the determinant $\sum \pm A_1 B_2 C_3 \dots$ does not vanish, then there are no non-zero values of $m_1, m_2, \dots, m_{h-1}$ for which all the equations

$$\begin{aligned} A_1 m_1 + A_2 m_2 + \dots + A_{h-1} m_{h-1} &= 0, \\ B_1 m_1 + B_2 m_2 + \dots + B_{h-1} m_{h-1} &= 0, \\ &\vdots \end{aligned}$$

would be satisfied. Thus it cannot be the case that

$$\Phi_1(\alpha)^{m_1} \Phi_2(\alpha)^{m_2} \dots \Phi_{h-1}(\alpha)^{m_{h-1}} = 1$$

for any integral values of $m_1, m_2, \dots, m_{h-1}$; for otherwise, by the irreducibility of the equation on which everything rests, it would follow that

$$\begin{aligned} \Phi_1(\alpha')^{m_1} \Phi_2(\alpha')^{m_2} \dots &= 1, \\ \Phi_1(\beta)^{m_1} \Phi_2(\beta)^{m_2} \dots &= 1, \\ \Phi_1(\beta')^{m_1} \Phi_2(\beta')^{m_2} \dots &= 1, \quad \text{etc.} \end{aligned}$$

Multiplying the equations two by two gives, for all h equations,

$$a_1^{m_1} a_2^{m_2} \dots = 1, \quad b_1^{m_1} b_2^{m_2} \dots = 1, \quad \text{etc.},$$

which is precisely impossible. Hence $\Phi_1(\alpha), \Phi_2(\alpha), \dots$ form a system of independent units.

I think such a reversal of the order of your conclusions would guard against certain misinterpretations. The words on page 106 and the following passage could lead one to believe, first, that for a system of independent units no single one of the above h equations could hold, whereas it is only impossible that all of them hold; and, secondly, that the equation $a_1^{m_1} a_2^{m_2} \dots = 1$ implies the equation $b_1^{m_1} b_2^{m_2} \dots = 1$, whereas only from the existence of

$$\Phi_1(\alpha)^{m_1} \Phi_2(\alpha)^{m_2} \dots = 1$$

does the existence of all the others follow. The equation

$$\Phi_1(\alpha)^{m_1} \Phi_1(\alpha')^{m_1} \Phi_2(\alpha)^{m_2} \Phi_2(\alpha')^{m_2} \dots = 1$$

gives no conclusion about the persistence of the same equation when α is changed into β .

I mention only the cases in which, for a system of independent units $\Phi_1(\alpha), \dots, \Phi_{h-1}(\alpha)$, one or more, though of course not all, of the equations

$$a_1^{m_1} a_2^{m_2} \dots = 1, \quad b_1^{m_1} b_2^{m_2} \dots = 1,$$

are in fact satisfied. If, for example, there exists a unit $\Psi(\alpha)$ whose analytic modulus is 1 but which is not a root of unity, then by your fundamental theorem one must have

$$\omega \Psi(\alpha) = \Phi_1(\alpha)^{m_1} \Phi_2(\alpha)^{m_2} \dots \Phi_{h-1}(\alpha)^{m_{h-1}},$$

where $\Phi_1(\alpha), \dots$ are the fundamental units and ω denotes a root of unity. If one changes $\sqrt{-1}$ into $-\sqrt{-1}$, this becomes

$$\omega^{-1} \Psi(\alpha') = \Phi_1(\alpha')^{m_1} \Phi_2(\alpha')^{m_2} \dots \Phi_{h-1}(\alpha')^{m_{h-1}},$$

and, multiplying and taking account of $\Psi(\alpha)\Psi(\alpha') = 1$, one obtains

$$1 = a_1^{m_1} a_2^{m_2} \dots a_{h-1}^{m_{h-1}},$$

where the m 's are non-zero.

There would be many more things to mention here, but they are not worth the writing. I leave this subject only to ask you whether you consider the observation interesting that one can find cyclometric solutions of

$$x^2 + Dy^2 = 4p$$

when $p \equiv 1 \pmod{D}$. This is easily seen as a generalisation of Jacobi's Ψ 's, from which only the solutions of

$$x^2 + 3y^2 = 4p, \quad x^2 + 7y^2 = 4p$$

result. The whole observation has merely fallen to me from more general considerations, which otherwise have not yet shown themselves sufficiently fruitful.

For greater simplicity take $D = \lambda$, a prime of the form $4n + 3$. If $\alpha^\lambda = 1$, $x^p = 1$, and $p = k\lambda + 1$, then in Jacobi's notation

$$(\alpha, x) = x + \alpha x^g + \alpha^2 x^{g^2} + \dots.$$

The product $\Pi(\alpha^a, x)$, extending over all quadratic residues a of λ , is easily seen to be of the form $\frac{1}{2}u + \frac{1}{2}v\sqrt{-\lambda}$, where u and v are integers. One has

$$\frac{u + v\sqrt{-\lambda}}{u - v\sqrt{-\lambda}} = \left(\frac{U + V\sqrt{-\lambda}}{U - V\sqrt{-\lambda}} \right)^{\frac{\sum b - \sum a}{\lambda}},$$

where U and V are the numbers in the solution of $U^2 + \lambda V^2 = 4p$. Thus for given p and λ one can set up an equation of degree

$$\frac{\sum b - \sum a}{\lambda},$$

which, if $U^2 + \lambda V^2 = 4p$ is solvable in integers, has a single rational root from which U and V follow immediately.

I have, to my alarm, again become far too expansive, and I therefore beg your pardon. I also ask you to commend me and my wife most warmly to your wife, and to preserve for me always your very precious goodwill. With most intimate attachment and reverence, your grateful pupil, Leopold Kronecker.

Kösen, 26 June 1856.

IV. Kronecker to Dirichlet

Most honoured Professor, I have just written a few lines to Dr. Dedekind, and my old attachment drives me also to send you some words, even at the risk that your interest in me may meanwhile have cooled somewhat. I do not really see a prospect of seeing you personally soon, for you always travel westward and, for many reasons, I cannot come to Göttingen.

My mathematical work has gone strangely this past winter. Both in my algebraic elaborations and in the further investigations on the affects of equations I was constantly interrupted by things apparently little connected with them, though I had no reason to be dissatisfied with these interruptions. I found many very interesting matters. Since I feel so independent that not the least trace of ambition troubles me, and since my joy lies solely in the recognition of what is true, it matters little to me on what exactly I spend my time, provided only that I use it well.

The two principal episodes of my work concerned the complex multiplication of elliptic functions and the theory of the most general ideal numbers. Regarding the first, I already communicated my first results to you here in December. In the first three months of this year I found many similar results, proved everything rigorously, and really completed the investigation in question. The little mathematical region explored in this connection was of the greatest interest to me in many respects: the elegance of the results, the simplicity of the methods of proof, the connection between analysis and number theory, and the further prospects opened by the matter—all this combined to spur my zeal in the investigation and afterwards to reward it.

I shall mention briefly the most essential results. First, the moduli k for which multiplication by $a + b\sqrt{-D}$ occurs are roots of a solvable equation of degree H , whose coefficients are complex numbers of the form $a + b\sqrt{-D}$, where H denotes the number of quadratic forms for determinant $-D$. Secondly, the equation has the character of the equations treated by Abel: all its roots are rational functions of one of them, and if $k, \theta(k), \theta_1(k)$ are such roots, then

$$\theta\theta_1(k) = \theta_1\theta(k).$$

The number of cycles or periods arising here according to Abel equals Gauss's exponent of irregularity for $-D$. If this number is 1, so that the equation is, as I call it, "Abelian," then the determinant is regular. I do not consider it hopeless that I shall still be able to determine whether that equation is Abelian or not, and hence whether the determinant is regular or irregular. For the moment I possess only this one criterion, but it is remarkable enough.

Thirdly, from the H moduli k one can express transcendently the coefficients of a system of non-equivalent quadratic forms of determinant $-D$. This result is all the more important since it could probably become a means of passing from the theory of elliptic functions to that of quadratic forms. Fourthly, in setting up the Abelian equations with complex multiplication I was led to transformations of theta-products and forms which in several cases give very short proofs and new forms of already known theorems.

Fifthly, I believe I have found a theorem on the number of forms for a real determinant in the complex theory of numbers. The formula is as follows. Let $F(m)$ be the number of forms for determinant $-m$. Then

$$F(n) + 2F(n-1) + 2F(n-4) + 2F(n-9) + 2F(n-16) + \cdots = \Phi(n),$$

where $\Phi(n)$ denotes the sum of those divisors of n which are greater than a certain bound indicated in the original computation, and where $n \equiv 3 \pmod{4}$. If n is prime, then $\Phi(n) = n$, and a noteworthy induction property appears for the determinants

$$n, \quad n-1, \quad n-4, \quad n-9, \quad n-16, \dots$$

For the reduced forms belonging to these determinants there are congruence relations modulo n , so that their coefficients, in the appropriate order, are congruent modulo n .

Sixth and seventh, the same circle of ideas gives relations among modular equations and form numbers, and in several cases the necessary computations amount essentially to reading off the terms. Eighth, the moduli k for which modular equations have equal roots are of the kind already described. Kronecker breaks off here, remarking that enough has been said and that these details must await a more orderly exposition. He adds that he is not yet ready to write on the composition of forms, and that he hopes above all to discuss the matter personally with Dirichlet.

With this he closes the mathematical part of the letter, asking Dirichlet to preserve his valued goodwill, sending greetings to Dirichlet's wife, and signing as his devoted pupil, Leopold Kronecker. Berlin, 17 May 1857, Köthener Strasse 12.

V. Dirichlet to Kronecker

Dr. Kronecker, Berlin, Köthener Strasse No. 12. Göttingen, 4 July 1857.

Since you know from experience, dear friend, how deeply rooted in me is the bad habit of answering even the letters of my dearest friends very late, you will hardly be surprised that the news brought here by Borchardt and Ernst, namely that you were absent from Berlin, prevented me from carrying out my firm resolve to write to you during the Pentecost holidays. But since, as I learn from Dedekind, you have returned and will soon leave Berlin again, I cannot delay longer in thanking you most warmly for your very interesting letter.

For the great pleasure caused by the communication of your beautiful discoveries, I can find no more suitable expression than to cry to you with full conviction, *macte virtute*. At the same time I cannot conceal that a little egoism is mixed with this pleasure, for in all modesty I cannot deny myself the testimony that I first introduced you into the lower regions of one of the sciences on whose heights you now stride as a master. I deliberately speak only of one of these sciences, for as regards your algebraic greatness I must declare myself wholly innocent.

My interest in the remarkable relations you have found among quadratic forms, algebra, and elliptic functions is all the livelier because I myself have sometimes traced similar relations. A point of view connected with this is indicated in my paper abbreviated "R. s. d. a.", paragraph 7; yet I later noticed that what is said there is much obscured by the condition assumed there, namely that in the double summations over x and y the trinomial

$$ax^2 + 2bxy + cy^2$$

should be relatively prime to D . I have followed the subject somewhat further from the point of view, much simplified by removal of that condition, but if I interpret correctly your unfortunately very brief communications, the remarks I made have

hardly anything in common with your beautiful results. Your results seem essentially to rest on specialising the modulus, while my consideration presupposes the indeterminacy of k and of the corresponding q .

I am therefore all the more eager to learn the foundation and proper starting point of your investigations. I believe I may count on their early publication, since you yourself describe the work as essentially finished and of moderate extent. As for myself, I am now engaged in working out the hydrodynamic paper whose subject we touched upon briefly at Christmas. The main result can be formulated approximately as follows.

If an incompressible homogeneous fluid mass, whose parts attract one another according to Newton's law and which at its surface is under a constant pressure, or a pressure depending only on time, initially has the shape of an ellipsoid; and if the initial motion is such that it can be decomposed into two simpler motions—a rotation about an axis through the centre with the same angular velocity for all mass elements, and a second motion in which all mass elements move, with respect to three mutually perpendicular planes through the centre, so that their velocities resolved perpendicular to each of these planes are proportional to their distances from them—then at any later time the surface will have the form of an ellipsoid concentric with the original one, whose axes, however, vary in length and direction with time. For the motion of the individual elements the same holds as was assumed at the first instant: at every time the instantaneous motion is composed of two simple motions of the kind just defined, but with the axis of rotation and the three corresponding planes generally variable with time.

It is easy to see that the assumed initial state of motion involves eight arbitrary quantities, and it is remarkable that for such an initial state the general character of the resulting motion can be specified. I say the general character, because for complete knowledge of the motion nine functions of time must be determined, defined by as many equations—one finite equation and eight differential equations of the second order—for which one can give generally seven first integrals. A complete solution can be carried out only in special cases; one of the simplest is when the ellipsoid has two equal axes and the motion begins without initial velocity.

From my equations follows the solution of a problem that seems to present some interest, even though the result is purely negative. The known results found by Maclaurin and Jacobi lead naturally to the question in what cases a fluid homogeneous ellipsoid, whose elements attract one another according to Newton's law, can rotate like a rigid body about its centre of gravity. The answer is that this is possible only when the motion takes place about a fixed axis coinciding with one of the principal axes of the ellipsoid; this is the case studied by Maclaurin and Jacobi.

The rest of the letter conveys family greetings and an apology from Dirichlet's wife for having been unable to visit Kronecker's wife during her short and hurried stay the previous winter. Dirichlet closes: *Vale et mihi fave*. Göttingen, 4 July 1857.

VI. Dirichlet to Kronecker

I hasten, dear friend, to thank you for your friendly letter and at the same time to send you, according to your wish, the earlier one from May. It goes without saying that you must later return the latter to me, since it is my property; but there is no haste, for I have at every moment a copy made by Dedekind. I take formal

note, as the jurists say, of your friendly promise to visit me, though I hope perhaps to see you even earlier in Berlin. As to your commission concerning Riemann, whose present whereabouts I do not know with certainty—he was for a long time in Harzburg—I discharge it by writing today to Dedekind, so that through him your request may reach Riemann.

Please recommend me and my wife most warmly to your wife, and greet our friends Ehlers, Kummer, Weierstrass, and Borchardt, if he has returned, from your faithfully devoted Dirichlet. Göttingen, 4 October 1857.

VII. Kronecker to Dirichlet

Most honoured Professor, first of all my heartiest thanks for the extremely prompt fulfilment of my recent request. Meanwhile I have been struck by a very severe loss: my mother, in the fifty-ninth year of her life, died so suddenly, though of a chronic illness, that when I hurried to Liegnitz six weeks ago at the first news, I no longer found her alive. I remained there several weeks with my poor father and was myself greatly shaken in mind and body by the blow. Afterwards there were several minor illnesses in my little family here. You can imagine that my work advanced only slowly, and I must ask your indulgence.

I take the liberty of enclosing the small note I have drafted on my “elliptic-function work” and ask that you look through it. Since spring I have altered the treatment in two respects. First, I have dropped for the time being the assertion about the irreducibility of the equations on which the modulus values depend, for my proof of it was false. I am still convinced that those equations are irreducible, but because I cannot yet prove it, I had to abandon for the time being the conclusion about the irregularity exponent founded on it. As long as that irreducibility is not established, the conclusion could be made only in an alternative form, and that is neither beautiful nor simple enough to be included in the enclosed note.

Secondly, in spring I had the mistaken opinion that I was, as it were, finished with the mathematical field under investigation, whereas now a fullness of new and rich material offers itself. Nevertheless I shall this time, contrary to my habit and inclination, proceed to publication before I have worked the matter through completely. In two parts of my work I shall give the solvability of those equations and their application to the theory of quadratic forms of negative determinant; the further applications, especially to the composition of forms, I shall probably postpone.

I hope you will approve this moderation in your poor pupil, for I finally see that I can no longer strive after my ideal: to produce, even in lower regions, a work as complete in itself as yours all are. It is my cross that I am always led into fields requiring better intellectual powers than nature has given me and more time than I can count on with my actually very unreliable body. In my exposition I shall presuppose nothing from the theory of quadratic forms, in order to show clearly that the elliptic functions make all this for themselves. But before I do the same for composition, I must absolutely speak with you, the master of all these regions.

I hope you will come here in the Christmas holidays; if not, I shall later come to Göttingen. Kummer, who has progressed very beautifully with his proof of the reciprocity law, asks me to convey his warmest greetings. Please commend me and my wife to your wife, who I hope has arrived there safely and who has not yet entirely overcome her influenza. I ask you yourself, honoured Professor, to

preserve for me the goodwill that I value above all. Berlin, 2 December 1857. Your faithful and grateful Leopold Kronecker. Please give Riemann and Dedekind my warm greetings.

VIII. Dirichlet to Kronecker

To Dr. Kronecker, from Berlin, presently in Ilsenburg. Göttingen, 23 July 1858.

Since you, dear friend, recently expressed at the Harzburg station the wish to learn the starting point of my investigations on the connection between quadratic forms of negative determinant and elliptic functions, I shall communicate my fundamental formula for the special case in which the determinant is a negative prime p of the form $4n + 3$, in a few words and as well as can be done without entering into lengthy discussions for which I lack leisure. I already indicated this connection in the second part of my paper "R. s. d. a. etc."; but the matter is complicated there by the fact that numbers having a common divisor with the determinant are excluded from the representation. If this assumption is removed, the formula for the special case just mentioned, using Jacobi's notation, is

$$\frac{k}{\sqrt{p}} \frac{K}{\pi} \sum \left(\frac{s}{p} \right) \sin \operatorname{am} \left(s \frac{4K}{\pi} \right) = \sum q^{\frac{1}{4}(ax^2+2bxy+cy^2)} + \text{etc.}$$

The sum on the left extends from $s = 1$ to $s = p - 1$, and the double sums on the right refer to all systems x, y for which the particular quadratic form (a, b, c) , of the first kind or properly primitive, is odd. The condition just stated could easily be removed if one also adjoined forms of the second kind. Each double sum can then, as already noted in the cited place, be decomposed into a finite number of products of two simple sums of the form

$$\sum_{x=-\infty}^{\infty} q^{(\delta x + \varepsilon)^2},$$

where δ and ε are rational numbers. Every such simple sum can, as follows easily from Jacobi's properties of ϑ , though to my knowledge this has not previously been observed, be brought into the form

$$\sqrt{\frac{K}{a}} \varphi(k),$$

where $\varphi(k)$ is an algebraic function of the modulus k belonging to q . Since, on the other hand, every $\sin \operatorname{am}$ occurring on the left is likewise an algebraic function of k , the equation expresses a relation among algebraic functions of the still variable k . One can therefore say that the whole theory of forms of negative determinant and of the numbers represented by them comes back to relations among roots of algebraic equations and must be derivable from those relations—a very noteworthy rapprochement, since no trace of such a connection has so far appeared for forms of positive determinant.

Since our recent conversation on the ride from Ilsenburg to Harzburg I have succeeded in narrowing substantially the estimate of the sum

$$\sum_{s=1}^n \left[\frac{n}{s} \right],$$

where the brackets denote, after Gauss, the greatest integer. Previously I could give it only with an error of order $\sqrt{n}$. The discovery of the method used for this, which will probably also apply to the following cases, gives me great pleasure, but it comes at an inconvenient time because it draws me away from completing the hydrodynamic paper, which must finally be finished.

Please recommend me to your wife and to the other ladies in Ilsenburg, as well as to Dr. Benzler. Göttingen, 31 July 1858. Your faithful Dirichlet.

Schering's note. In July Dirichlet visited Kronecker, who spent the summer in Ilsenburg, for several days. On Dirichlet's return journey to Göttingen Kronecker accompanied him by carriage from Ilsenburg to Harzburg.

IX. Dirichlet to Kronecker

Göttingen, 31 July 1858. Since I wrote to you, dear friend, yesterday week, I have received from my wife in Prussia a letter from which it seems to follow that she did not see Borchardt, who had expressed the intention of visiting her in Charlottenburg. I must therefore fear that the condition of our friend has again worsened, and I ask you to tell me in two words what you have learned about him from Berlin. A letter from Heine received yesterday does not mention with a syllable that you had summoned him to the Harz; it will probably now not take place at all, since your departure for Heligoland, as you recently told me, must be imminent. With my best recommendations to your wife, your most devoted Dirichlet.

X. Kronecker to Dirichlet

Ilsenburg, Sunday 1 August 1858. Most honoured Professor, I have just received your letter of yesterday and hasten to give you news of our friend Borchardt, with whom I was in Braunschweig a week ago today. How that happened without my being able to fulfil the arrangement made with you I must explain in a few words. The fairly persistent fine weather on the Monday after your departure from here impelled me to carry out my plan of making a several-day excursion to the Brocken and the Lower Harz. Weather and other disturbances prolonged my absence until Saturday a week ago. In the meantime I had written neither to Borchardt nor to Heine, since according to Kummer's earlier communication on Borchardt's health I regarded his departure as still postponed for some time.

To my astonishment, on my return here Saturday I found Borchardt's notice that he would arrive in Braunschweig at noon on Sunday. It was therefore quite impossible to inform you in time, especially since I knew that you would again lecture on Monday and that even a longer waiting by Borchardt in Braunschweig would not have helped. I therefore travelled over alone on Sunday and was with Borchardt from one to eight o'clock. He naturally regretted very much being deprived—perhaps also of Heine's visit—by circumstances whose prevention was perhaps partly within my power. You now have all the data, honoured Professor, to pronounce the verdict on me yourself.

So much for my excuses. As for Borchardt, I found him admittedly very weak and impaired; nevertheless I hope that he will have completed happily the journey to the Isle of Wight, which he made in short daily stages in the company of an experienced servant. He expects to meet on the Isle of Wight his cousin Helborn, now

married Sussmann, with her husband, so that he will feel all the less abandoned there. Borchardt had, as I heard from Berlin a few days ago, had another small attack a few weeks earlier, and from that time dates the real improvement, whereas until then the illness had run a more creeping course but had also never really yielded. Borchardt plans to stay about five weeks in England and, if he then feels well, to go for a time to Sorrento. The two sea coasts lie far apart, but railways and steamships run quickly.

I shall learn Borchardt's address on the Isle of Wight only in Heligoland, where I intend to travel the day after tomorrow with my son Ernst, and where, to judge from a Berlin letter received yesterday, Weierstrass and Kummer may perhaps follow me.

After all these personal matters I come now to thank you most warmly for your friendly letter of 23 July and to tell you how flattered I feel by this prompt communication, and still more how greatly the content interested me. That you have advanced so far in determining

$$\sum \left[\frac{n}{s} \right]$$

is splendid indeed, and no consideration should lessen your joy in it. Certainly no hesitation will keep you from the further pursuit of these investigations, for the force of their attraction is too great. Your comprehensive mind will surely have no difficulty working out finished investigations alongside work in quite different fields. That is precisely the advantage of our great men: they not only possess an abundance of ideas, but the abundance does not hinder them from exploiting those ideas.

Your beautiful communications about elliptic functions I shall use properly as soon as I return to Berlin. I shall try to obtain the transformation you indicated,

$$\sum q^{\frac{1}{4}(ax^2+2bxy+cy^2)}$$

into products of two simple sums

$$\sum q^{(\delta x + \varepsilon)^2},$$

and especially to determine the rational numbers δ and ε . These probably depend directly on the coefficients of the forms, at least in such a way that-while the left side of your principal equation evidently contains algebraic functions depending on the modular equation of order p -the right side contains algebraic functions arising from modular equations of other orders. Such relations are of the highest algebraic interest to me. Known examples already exist, but those resulting from your equation are certainly quite different from the known ones. The number-theoretic consequences of your equation must likewise be very different from those that I drew from the theory of elliptic functions. I believe that your equation will yield interesting results also when one gives the modulus k one of the special numerical values I have investigated.

All my ideas and intentions in this direction are naturally still very indefinite, for Ilseburg is not a good climate for mathematical studies. But from Berlin I hope to show my gratitude for your valuable communications. My dear wife asks to be warmly recommended to you, and I myself ask you to preserve for me your very precious friendship. Your faithfully devoted Kronecker.

Best greetings to your son Ernst and my best recommendations to your mother, as well as to Weber, Listing, and Riemann. I ask you to say nothing to anyone

else for the moment about the ideas I have communicated to you concerning the algebra of the future, especially concerning the use of Jacobi's linear equations for $\sqrt{n}$. I should first like to see whether my speculations succeed.